Leakage-free, explainable Parkinson's disease screening

Gait, wrist-sensor and voice models with subject-grouped evaluation - project overview, literature and plan

At a glance

- **Problem.** Machine-learning papers report very high accuracy for Parkinson's disease (PD) from gait, but the standard way of splitting data can let the same person appear in both training and test sets.
- **Our claim.** Measured on the same public dataset, trial-level cross-validation inflates accuracy by roughly 15-30 points for 4-class staging; a subject-grouped protocol gives the honest number.
- **Our contribution.** (1) a quantified, reproducible leakage measurement; (2) richer gait features under honest evaluation; (3) the same test repeated on voice and wrist data; (4) SHAP explanations a clinician can read; (5) an exploratory fusion demo.
- **Status.** Gait pipeline and leakage measurement are done and pushed to GitHub. Voice, wrist, SHAP, fusion and the paper text are still to do.

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1. Why this project and who it helps

Parkinson's disease is a chronic neurodegenerative disorder. Clinical severity is rated with scales such as the Hoehn and Yahr (H&Y) stage and the UPDRS, which depend on the examiner's expertise and are partly subjective. Gait abnormality is an early, measurable sign, so foot-pressure sensors are an attractive route to objective, repeatable screening.

Most published classifiers evaluate themselves in a way that can overstate how well they will work on a new patient. This project shows how large that effect is on the well-known PhysioNet gait dataset, fixes the evaluation, and asks whether an honest model can still be useful and explainable.

Objectives

- **Re-implement** the baseline pipeline (Balaji et al. 2020) on PhysioNet gaitpdb and evaluate it under both protocols. Our trial-level numbers do not match the baseline's ~99% headline; we say so plainly.
- **Quantify leakage** with a reproducible script (5 seeds, in-fold preprocessing check, leak audit).
- **Improve features** (centre of pressure, variability, symmetry; frequency, regularity and demographics planned) and measure gains honestly.
- **Generalize**: repeat the leaky-vs-grouped test on voice (UCI speech) and wrist data (PADS).
- **Explain** predictions with SHAP in clinically named features.
- **Demo** late fusion of three models into one risk score (exploratory, not validated).

Audience	What they get
ML / biomedical researchers	A warning and a fix: reported PD accuracies (often 90%+) may be inflated by leakage; here is a protocol that avoids it. This is the main contribution.
Neurologists and clinicians	Realistic performance expectations and feature-level explanations they can check against clinical knowledge (stride variability, asymmetry, reduced force).
Patients and screening programs (indirect)	Honest models are the only ones that could safely support low-cost, objective, repeatable screening where specialists are scarce.
Reviewers and journals	A reproducible pipeline: public data, fixed seed (42), numbered scripts, results that trace to CSV files.

What this is not

- Not a diagnostic tool - a research study of screening support.
- No claim about fused accuracy across the three modalities: no public dataset has all three on the same people.
- 4-class staging is weak, especially for the Severe class (10 subjects); the paper must say so.

2. Spotlight: the baseline research paper

Balaji E., Brindha D., Balakrishnan R. (2020). Supervised machine learning based gait classification system for early detection and stage classification of Parkinson's disease. *Applied Soft Computing* 94, 106494.

- **Goal:** classify PD severity stages from foot-pressure (VGRF) gait data instead of relying on subjective clinical rating.
- **Data:** PhysioNet gait dataset - 73 healthy controls and 93 PD patients (H&Y 2: 55, 2.5: 28, 3: 10); 8 sensors per foot, 100 Hz, ~2 minutes of walking.
- **Method:** first and last two gait cycles discarded; statistical and kinematic (spatiotemporal) analyses; Shapiro-Wilk normality test; Pearson-rank correlation feature selection.
- **Classifiers:** Decision Tree (DT), SVM, Ensemble (bagged trees, 30 learners) and Bayes; 10-fold cross-validation, averaged over 10 independent runs; accuracy, sensitivity, specificity, PPV, NPV, F-score, confusion matrices and ROC.
- **Headline:** up to 99.4% accuracy (DT) for stage classification, and the best result among the methods compared in its Table 9.

Results reported by the baseline

Analysis	Decision Tree	SVM	Ensemble	Bayes
Statistical features (text)	99.4%	97.6%	95.1%	69.7%
Kinematic features (Table 8)	99.4%	99.4%	95.2%	69.9%
Ours: subject-grouped, 4-class	43.2%	42.8%	48.8%	38.8%

Baseline figures are as printed in the paper. Ours are 10-fold subject-grouped CV (seed 42) from the same four classifier families.

What we adopt, and what we question

We adopt from the baseline	We question / extend
The public VGRF dataset and the 4-stage H&Y framing (Healthy / Mild / Moderate / Severe).	The paper describes 10-fold CV repeated 10 times but does not describe grouping folds by subject. If folds were drawn at the trial level, one person's trials can sit in both training and test data, which can inflate accuracy. This is our hypothesis; our experiments measure how large the effect is on this same dataset.
Statistical plus kinematic features, correlation-based selection, and the DT / SVM / Ensemble / Bayes lineup.	Step length, stride length and speed need distance data that the public VGRF files do not contain, so we cannot reproduce those features. We add centre-of-pressure, stride-time variability and symmetry features instead.
Reporting accuracy, sensitivity, specificity, PPV, NPV, F-score and AUC per stage.	Accuracy alone can hide an unlearnable minority class (Severe, 10 subjects); we also report macro-F1 and per-class metrics. We also add explanations (SHAP), which the paper does not provide.
Their own stated limitations: motor symptoms only, and only supervised classifiers.	They suggest adding non-motor and tremor data. Our wrist-sensor (PADS) and voice branches follow that direction, with honest, subject-grouped evaluation.

Note: the baseline text cites 279 recordings; our copy of the dataset has 306 trial files, all of which are used. The reason for the difference should be confirmed before the paper is written.

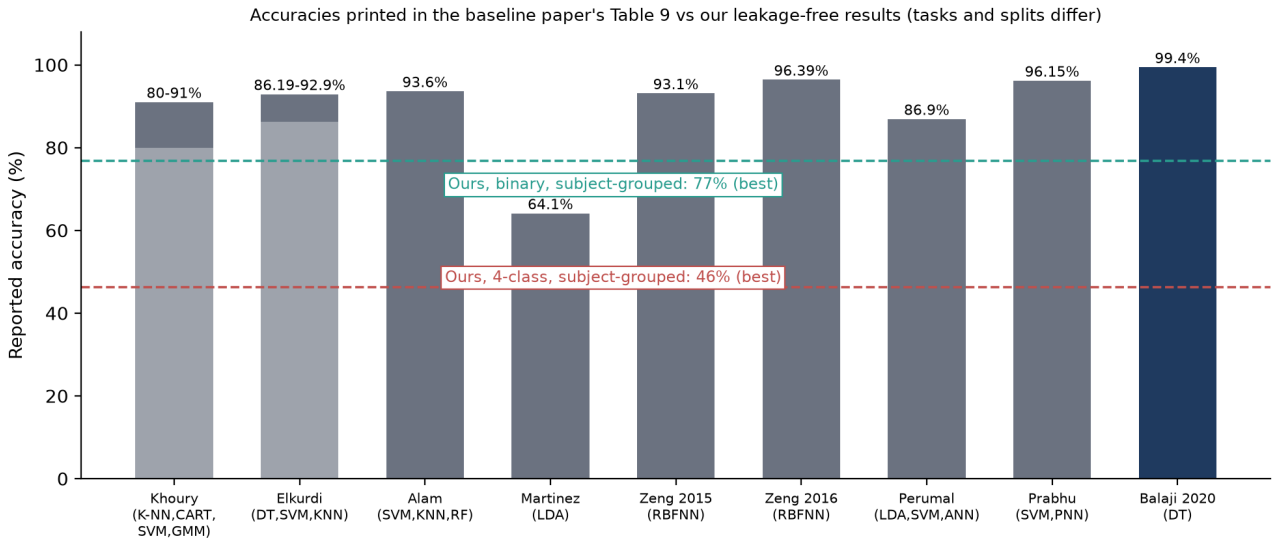
3. Literature review and the research gap

Related work on PD detection from the same gait dataset, as summarised in the baseline paper (Balaji et al. 2020, Section 2 and Table 9). Details below are taken from that paper's own description of each work; consult the primary sources before citing.

3.1 Machine-learning gait work on PhysioNet VGRF data

Work (as cited in the baseline)	Approach	Reported outcome / note
Lee and Lim (2012)	Wavelet-based features (40 features from statistics and frequency distribution); neural network with weighted fuzzy membership functions	Max accuracy 81.63%; many wavelet features raise computational cost
Asuroglu et al. (2018)	Locally weighted random forest regression; 16 time-domain and 7 frequency-domain features; models GRF to PD symptoms	Quantitative motor-symptom assessment; statistical GRF only (weight-dependent), no kinematic analysis or severity levels
Zhao et al. (2018)	Two-channel hybrid of LSTM and CNN learning spatiotemporal patterns from VGRF	Significant accuracy gain from deep learning; suggests fusing other sources (MRI, biochemical)
Prabhu et al. (2018)	Recurrence quantification analysis (RQA) plus statistics; hill-climbing feature selection; PNN and SVM	Works even for short gait series; 96.15% in Table 9
Figueiredo et al. (2018)	Review of ML for gait pattern recognition; kernel PCA and genetic algorithm for dimension reduction	Comparative analysis; uses cross-validation to assess classifiers
El Maachi et al. (2020)	Deep 1D-ConvNet for PD detection and severity prediction from gait	Cited in the baseline as related deep-learning work
Perumal and Sankar (2016)	Gait and tremor assessment with wearable sensors; LDA, SVM, ANN	86.9% in Table 9

3.2 Accuracies compared in the baseline's Table 9



Bars show accuracies as printed in Table 9 (ranges where the paper gives a range). The two dashed lines are our best subject-grouped results. The tasks (binary vs multi-class), features and split protocols differ across papers, so this is context, not a like-for-like ranking.

3.3 Evaluation leakage in clinical machine learning (added by us - verify before citing)

A wider literature warns that record-wise cross-validation is over-optimistic when several records come from one person, and that data leakage is a common cause of unreproducible ML results. We cite these as the methodological grounding for our protocol:

- **Saeb et al. (2017)**, *GigaScience* - showed that record-wise cross-validation overestimates accuracy for smartphone activity and health data compared with subject-wise splitting.
- **Little et al. (2017)**, *GigaScience* - response and perspective on cross-validation strategies for subject-level data.
- **Kapoor and Narayanan (2023)**, *Patterns* - survey of data-leakage errors across ML-based science.
- **Little et al. (2009)**, *IEEE Trans. Biomed. Eng.* - PD voice dysphonia measures; the popular UCI voice data derive from this line of work and contain several recordings per person, so the same leakage risk applies.

3.4 Other modalities we use

Modality	Dataset and source	Why it matters here
Wrist tremor and hand movement	PADS smartwatch dataset (PhysioNet; npj Parkinson's Disease, 2024): 469 participants, accelerometer and gyroscope at 100 Hz, 11 tasks, includes differential diagnoses	Tremor and bradykinesia are core clinical signs; large, open and with subject IDs
Voice	UCI Parkinson's Speech with Multiple Types of Sound Recordings (Sakar et al., 2013): 40 training subjects x 26 recordings	Many recordings per subject makes it ideal to demonstrate recording-level leakage; small, so results are indicative
Explainability	SHAP (Lundberg and Lee, 2017)	Lets clinicians see which features drive a prediction

3.5 Positioning and the research gap

	Typical prior gait-PD work	This project
Evaluation split	Often trial-level CV; grouping by subject not stated	Subject-grouped CV as primary; leaky CV run alongside to measure the gap
Reporting	Single accuracy, one run	5 seeds, fold variance, per-class and macro-F1, in-fold preprocessing check, leak audit
Interpretability	Rarely explained	SHAP, clinically named features, per-patient report
Generalization	One dataset	Same leakage test on gait, voice and wrist data
Stated honesty	Headline accuracy	Null and weak results (Severe class, 4-class) reported openly

Research gap in one sentence

- Published gait-based PD classifiers report very high accuracy under evaluations that may leak subjects between training and test data; no one has measured that inflation on the standard dataset with a reproducible, subject-grouped protocol, shown it generalizes to other modalities, and made the honest models explainable.

4. Datasets

Model	Dataset	Subjects	Signal	Access	Status
A: Gait	PhysioNet gaitpdb v1.0.0	166 documented (93 PD, 73 healthy); 165 IDs in our feature file; 306 trials; cohorts Ga, Ju, Si	Foot VGRF, 16 sensors + 2 totals, 100 Hz; H&Y stages	Open	In use
B: Wrist	PADS smartwatch (PhysioNet)	469 (PD, differential diagnosis, healthy)	Wrist accelerometer + gyroscope, 100 Hz, 11 movement tasks	Open, CC BY-NC-SA 4.0, ~1.4 GB	Planned - download and check group counts
C: Voice	UCI Parkinson's Speech #301	40 train (20 PD, 20 healthy) x 26 recordings; 28 PD-only test	26 provided acoustic features, no raw audio	Open, CC BY 4.0	Planned

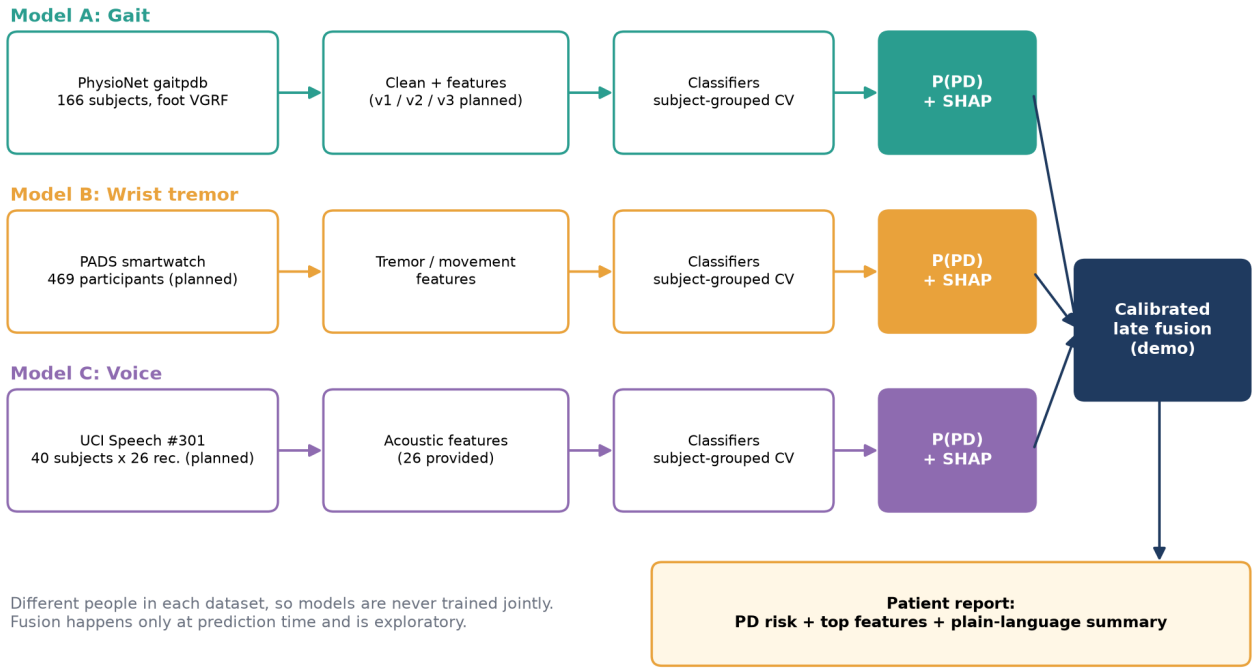
Gait class distribution (subjects)

Healthy	Mild (H&Y 2)	Moderate (H&Y 2.5)	Severe (H&Y 3)
73	55	28	10 (24 trials)

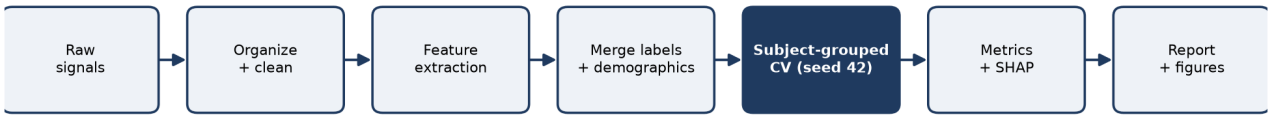
- The three datasets contain **different people**: rows are never merged, and models are never trained jointly.
- Only gaitpdb has H&Y staging. The wrist and voice models are binary (PD vs healthy), so the fused score is a PD-risk score.
- **Step length, stride length and gait speed** cannot be computed: the VGRF files have no distance data, and values cannot be borrowed from other datasets because they describe other people. Height-based proxies may be added only if labelled as proxies.
- The UCI test set contains PD patients only, so accuracy is computed on the 40-subject training set with grouped CV; the voice study is small and is presented as indicative.
- Dropped: handwriting/spiral and eye/pupil modalities; face deferred (no large public data).

5. System architecture and pipeline

Three independent models, each trained on its own public dataset with subject-grouped cross-validation and explained with SHAP. They are combined only at prediction time, on one patient's three recordings, by calibrated late fusion.



Per-modality pipeline



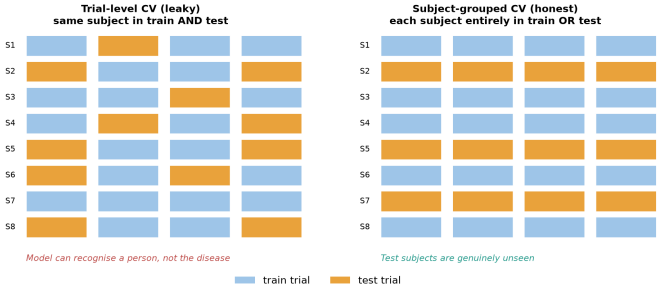
Preprocessing, feature selection and augmentation are fit inside training folds only.

Evaluation rules that apply everywhere

- Primary protocol: **StratifiedGroupKFold** by subject, 10 folds (capped by the smallest class), seed 42.
- Feature selection, scaling, SMOTE and augmentation are fit inside training folds only.
- Windows or recordings from one subject always stay in the same fold.
- Report accuracy, macro-F1, sensitivity, specificity, AUC, with fold variance.
- Fusion: calibrate each model (Platt or isotonic on validation data), then a simple or validation-AUC weighted average; never tune on test data; no fused-accuracy claim.

6. The leakage problem, explained and measured

Each person walked several times, so the dataset contains several trials per person. If trials are split at random, a person's other trials sit in the training set while one trial is tested. The model can succeed by recognising the person's walking signature rather than by recognising the disease. Subject-grouped splitting removes this shortcut.



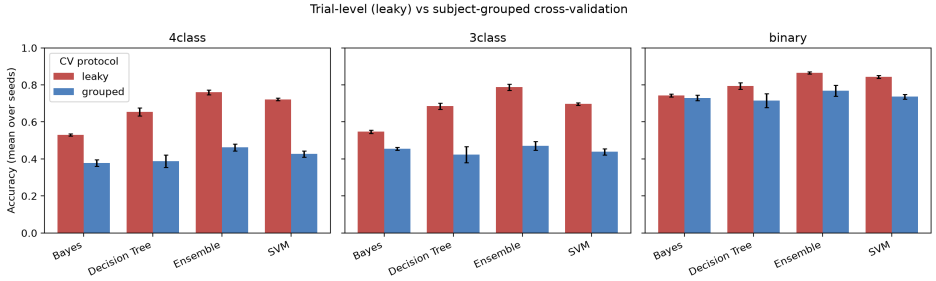
Measured with `scripts/14_leakage_comparison.py`: same features, labels, models and metrics; only the splitter changes. Five seeds. On average **63%** of the test trials in a trial-level fold belong to a subject who is also in that fold's training set.

Key takeaways

- 4-class staging: trial-level 53-76% vs subject-grouped 38-46%; the gap is 15-30 points, about 30 for Bagging and SVM.
- Binary PD vs healthy: the gap is much smaller (1-11 points), because two-class grouped accuracy is already high. Leakage matters most for fine-grained staging.
- The multi-seed grouped 4-class best result is 46% (Bagging); 48.8% is the single-seed (seed 42) figure.
- Our old working figure of "~67-70% vs ~40%" had no script behind it; it has been replaced by these measured numbers.

Scheme	Classifier	Trial-level (leaky)	Subject-grouped	Gap (points)
4-class	Decision Tree	65.4%	38.8%	26.6
	SVM	72.2%	42.7%	29.5
	Ensemble (Bagging)	76.0%	46.2%	29.8
	Bayes (GaussianNB)	53.0%	37.9%	15.1
3-class	Decision Tree	68.5%	42.3%	26.2
	SVM	69.7%	43.8%	25.8
	Ensemble (Bagging)	78.8%	47.1%	31.6
	Bayes (GaussianNB)	54.7%	45.5%	9.2
Binary	Decision Tree	79.3%	71.5%	7.8
	SVM	84.4%	73.7%	10.7
	Ensemble (Bagging)	86.5%	76.8%	9.6
	Bayes (GaussianNB)	74.4%	73.0%	1.4

Mean accuracy over 5 seeds, global preprocessing. Fitting preprocessing inside each fold gives the same picture (see leakage_comparison.csv).



7. What is done: results so far

Stage	Work	Script / output
Setup	Repo, .gitignore, data README, pushed to GitHub	README.md, data-README.md
Cleaning	Organize by cohort and group; drop NaNs and duplicates; trim edge cycles; validate format	01_organize_raw.py, 02_clean_signals.py, check_cleaned_data.py
Features	v1 statistics; v2 adds skew, kurtosis, IQR, centre of pressure, stride-time CV, symmetry, hysteresis stance/swing	03_extract_features.py, 03_extract_features_v2.py
Labels	H&Y and stage merged with features	04_merge_labels.py, subject_labels.csv
Baseline models	DT, SVM, Bagging, Gaussian NB, subject-grouped 10-fold	05_train_classifiers.py, 06_diagnose_errors.py
Label schemes	4-class, 3-class (Moderate + Severe merged), binary	07_train_multi_scheme.py
Full metrics	TP/TN/FP/FN, sensitivity, specificity, PPV, NPV, F-score, AUC per class	08_full_metrics.py
Class imbalance	Class weights and SMOTE - no meaningful gain for Severe	09_class_imbalance.py
Cohort analysis	Per-cohort and leave-one-cohort-out; Ju investigation	10_per_cohort_analysis.py, 11_investigate_ju_cohort.py
Leakage	Leaky vs grouped, 5 seeds, in-fold check, leak audit, figure	14_leakage_comparison.py

Subject-grouped results (10-fold, seed 42)

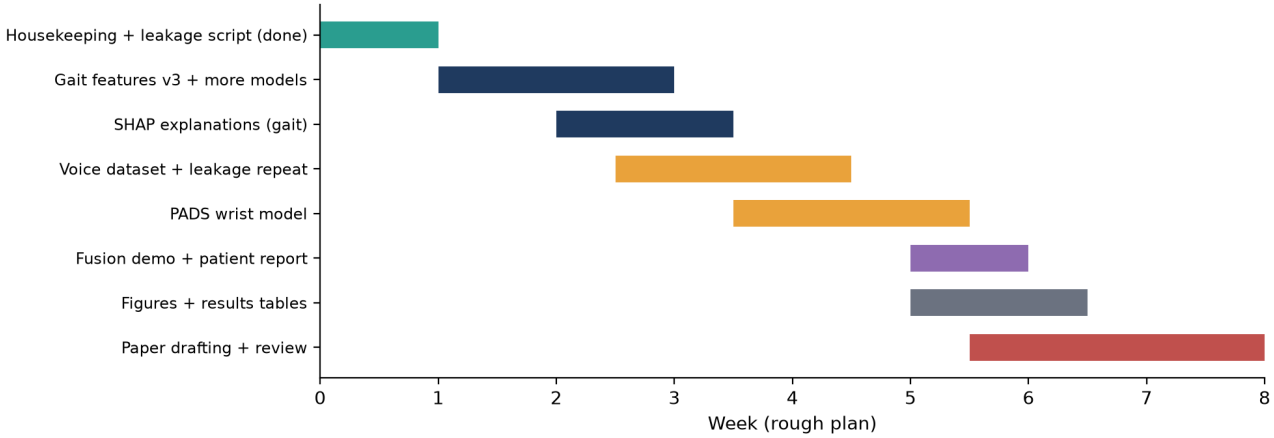
Scheme	Decision Tree	SVM	Ensemble (Bagging)	Bayes
4-class	43.2%	42.8%	48.8%	38.8%
3-class	41.2%	45.8%	46.4%	45.4%
Binary	68.8%	74.8%	77.1%	74.4%

Per-cohort behaviour (pooled 4-class model)

Cohort	Test trials	Accuracy	Macro-F1	Leave-one-cohort-out accuracy
Ga	113	46.9%	0.346	38.9%
Ju	129	31.8%	0.264	44.2%
Si	64	62.5%	0.502	48.4%

The Ju cohort is the weakest when pooled; when Ju is held out entirely, accuracy rises to 44%, so its low pooled score is not simply due to being unlearnable. The Ju investigation output has not yet been reviewed.

8. What remains: plan and timeline



Track	Tasks	Deliverable
Gait (Model A)	Features v3: demographics (report with and without - age confounds), FFT and wavelet, regularity (autocorrelation, sample entropy, DFA), double-support time. Add Random Forest, XGBoost, logistic regression, stacking. Optional sliding-window LSTM with augmentation in training folds only.	Improved honest accuracy; model comparison table
Explainability	SHAP TreeExplainer per model; clinically named feature groups; global importance and per-patient waterfall plots; plain-language summary template.	SHAP figures and patient report template
Voice (Model C)	Load UCI features; recording-level vs subject-grouped CV; models; SHAP.	Second leakage demonstration
Wrist (Model B)	Download PADS, confirm group counts, extract tremor-band (4-6 Hz) power, amplitude, variability and regularity features; PD vs healthy (optional PD vs differential dx).	Third leakage demonstration
Fusion demo	Calibrate, average, produce combined report for a demo patient.	Demo figure; no accuracy claim
Paper	Methods, results, discussion, limitations, then introduction and abstract; choose a venue; similarity check.	Manuscript draft

Immediate next steps

- Gait features v3 plus new models and SHAP (core of the paper).
- Download PADS and read its metadata for group counts; start any access requests early.
- Review the Ju cohort output; find the missing 166th subject; compare our pipeline with the baseline's tables.

9. Explainability and the expected patient output

For a patient's gait, wrist and voice recordings, the system reports the estimated chance of Parkinson's disease from each model and a fused overall score, plus the features that raised or lowered that estimate. Feature attribution uses SHAP; features are named in clinical terms (for example stride-time variability, left-right asymmetry, stance ratio, centre-of-pressure sway, tremor-band power, jitter and shimmer) so a clinician can judge whether the reasoning is plausible.

Mock-up of the report (numbers are illustrative only, not results)

	Gait	Wrist	Voice	Fused
PD risk (illustrative)	0.72	0.61	0.55	0.66

Top drivers (gait, illustrative)	Direction	Clinical reading
Stride-time variability	raises risk	Irregular walking rhythm, a known PD marker
Left-right asymmetry	raises risk	One side loaded differently from the other
Stance/swing ratio	raises risk	Altered timing of the gait cycle
Total force level	lowers risk	Within the typical range

Summary line

- "Elevated PD risk, driven mainly by high stride variability and asymmetric stance." This is a research screening estimate, not a diagnosis.

Cautions

- SHAP shows what the model used, not the disease mechanism; sanity-check that top features are clinically sensible (an odd feature can signal a data artefact).
- Per-modality accuracy is validated; fused accuracy is not, because no paired data exist.
- Weights for fusion come from each model's validation performance, never from test data.

10. Limitations, risks and decisions

Risk or limitation	Mitigation
Small sample: 166 subjects; Severe class only 10	Report macro-F1 and per-class results; keep 3-class and binary views; state the limitation
Extra rows (windows, augmentation) do not add subjects	Group by subject; only external datasets add real subjects, and only with shared features and binary labels
More features can overfit 166 subjects	Nested feature selection inside folds; regularized models; report fold variance
Voice set is small (40 subjects)	Present as indicative; use it to show leakage, not to claim state-of-the-art accuracy
No usable data for a modality (e.g. PADS counts differ)	Drop or reduce the branch and say so
Scope creep dilutes the thesis	Leakage claim first in every section; fusion stays a demo
Public repo includes the baseline PDF	Confirm copyright before keeping docs/paper.pdf public

Key decisions

- Thesis: subject-level leakage plus corrected evaluation plus richer features; gait paper stays leakage-first.
- Subject-grouped CV, seed 42; leakage quantified by script 14 rather than by memory.
- Datasets: gaitpdb, PADS, UCI speech. Handwriting and eye modalities dropped; face deferred.
- The alpha-synuclein aggregation idea (a tool called Kanya from Vercula) is excluded from this paper: the tool could not be verified, and it is a different research question. If pursued, it should be a separate study using established aggregation predictors and validation against published experiments.

11. References

A. Cited in or taken from the baseline paper and the dataset pages

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B. Added by us from general knowledge - verify authors, venues and details before citing

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Repository: github.com/devanshi1308/parkinsons - see prd.md, architecture.md, implementation.md and rules.md for the full plan and decision log.